2	(a)		Gravitational potential energy is taken to be zero at infinity and gravitational forces are attractive. Since the external agent is applying a force in the opposite direction relative to the displacement made by the mass, negative work is done by the external agent.	[B1] [B1]
	(b)	(i)	At infinity, kinetic energy and gravitational potential energy of probe is zero. By conservation of energy, the minimum kinetic energy required to for probe from surface to reach infinity is GMm/R_E Since E is smaller than minimum required Kinetic energy, the probe cannot travel to deep space. Thought Process: - Calculate/determine the minimum energy required for probe to travel to infinity from surface. - Compare the minimum energy required with the energy of probe. Comment Note: At surface, KE + GPE = 0 (Assume very far) By COE, Energy on Earth = Energy at infinity $KE_s + GPE_s = 0$ $KEs = -GPE = GMm/R_E$	[M1] [A1]
		(ii)	By conservation of energy Energy at surface = Energy at that point Initial KE + Initial GPE = final GPE + final KE $\frac{3}{4} GMm/R_E + (-GMm/R_E) = (-GMm/R) + 0$ $-\frac{1}{4R_E} = -\frac{1}{R}$ $R = 25.7 \times 10^6$ m OR Loss in KE = Gain in GPE $\frac{3GMm}{4R_E} = \frac{GMm}{R_E} - \frac{GMm}{R}$, note $R_E < R$ $R = 25.7 \times 10^6$ m	[C1] [B1] [A1]
		(iii)	Earth rotate in the West to East direction. Space probe launch in this direction will have an initial velocity, thus it has some initial KE, and not zero.	[B1] [A1]